

Can Today's Quantum Hardware Beat Random Guessing?

A head-to-head benchmark of five QPUs on a real optimization problem.

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Date March 2026

The Short Version

I ran the same optimization problem on five different quantum processors (four via Amazon Braket, one via IBM Quantum) and compared every result against random guessing. No error mitigation, no vendor-specific tricks. Just raw hardware output versus coin flips.

- On a perfect simulator the algorithm clearly beats random.
- Most hardware backends could not maintain that advantage once circuits got large.
- IBM Marrakesh was the exception, it kept improving as the problem scaled up to 61 qubits.

1 Setup

The task is feature selection on the German Credit dataset, encoded as a QAOA optimization problem. The full dataset produces a 61-qubit circuit after encoding. Each quantum circuit proposes a subset of features; each subset gets an energy score. Lower energy = better subset. The test: does the quantum output land at lower energies than 1,000 random guesses?

Every backend ran the same circuit with the same parameters. No error mitigation was applied anywhere, so the comparison is apples-to-apples.

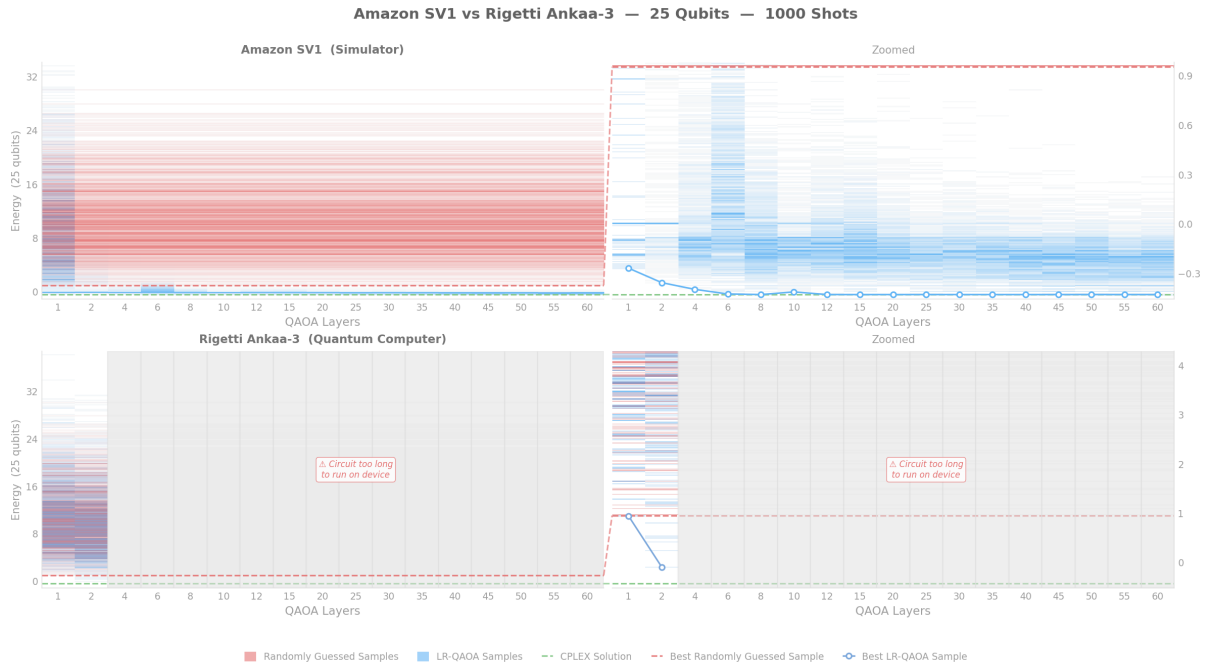


Figure 1: 25-qubit example. Top: ideal simulator (SV1). Bottom: Rigetti Ankaa-3. Blue = quantum output, red = random baseline, green dashed = exact optimum.

How to read this. Each stripe is one sample out of 1,000. Blue stripes below red stripes = quantum is beating random. The right panel zooms into the low-energy region where it matters most. On the simulator the blue distribution sits clearly below random. On Rigetti hardware that gap mostly disappears.



Figure 2: Full results across all backends and problem sizes. The Amazon Braket backends have much less data because they could not run many of the larger/deeper circuits.

2 Results

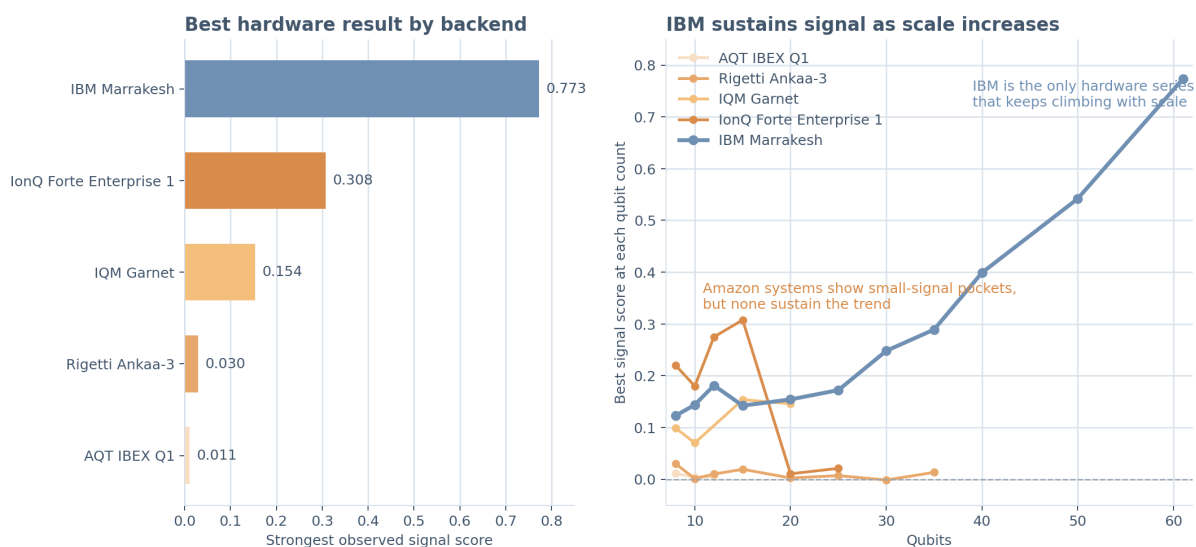


Figure 3: Signal vs. qubit count for each backend. Orange = Amazon Braket hardware, blue = IBM

Marrakesh.

The y-axis, what I call “signal” is the Jensen–Shannon divergence between the quantum and random energy distributions. A single number that captures two things: (1) how different is the quantum output from random, and (2) is that difference actually better, not just different? A score near zero means the hardware is doing no better than guessing. Positive means the quantum output is finding lower-energy solutions. Negative means it is doing worse than random. We define signal as follows:

$$\text{signal}(R, Q) = \text{JSD}(R||Q) \cdot \tanh\left(\frac{\mu_{\text{random}} - \mu_{\text{quantum}}}{\sigma_{\text{random}}}\right)$$

The first factor measures distributional distance, the second factor gives it a direction.

What the plot shows

The Amazon hardware lines (orange) either plateau near zero or drop off as qubits increase. Some stop entirely because the circuit exceeded what the device could run. The IBM Marrakesh line (blue) keeps climbing. It does not match the simulator, but it is the only hardware backend in this test whose output keeps separating from random as the problem grows.

Takeaway: On this workload, IBM Marrakesh was the only tested QPU that showed a real scaling trend rather than a small-problem demo effect.

3 Cost

Backend	Jobs	Pricing	Total cost	Cost / job
 IQM Garnet	32	\$0.00145/shot	\$56.00	\$1.75
 Rigetti Ankaa-3	33	\$0.00090/shot	\$39.60	\$1.20
 AQT IBEX Q1	11	\$0.02350/shot	\$261.80	\$23.80
 IonQ Forte Enterprise 1	16	\$0.08000/shot	\$1,284.80	\$80.30
Amazon Braket Total	92		\$1,642.20	\$17.85
 IBM Marrakesh	221 437 sec	\$1.60/sec	\$699.20	\$3.16

Figure 4: Cost comparison. All IBM pricing is pay-as-you-go equivalent for fair comparison.

IBM Marrakesh delivered the best results in the benchmark at less than half the total cost of the Amazon Braket runs combined. The IBM runs also fit inside the monthly free tier, so the actual out-of-pocket cost was \$0.

4 Bottom Line

Five quantum processors, one optimization problem, no error mitigation. Most hardware produced output indistinguishable from random guessing once circuits got large enough to matter. IBM Marrakesh was the clear outlier: it maintained a measurable advantage over random up to 61 qubits and did so at lower cost. On this workload, not all QPUs are equal, and only one showed credible scaling.